

Approach Document – SHL Assessment Recommendation Assistant

1. Project Overview

The objective of this project is to develop an AI-powered assistant that recommends appropriate SHL assessments from natural language hiring requirements. Recruiters often describe hiring needs conversationally by specifying roles, technical skills, experience levels, or hiring objectives rather than searching through assessment catalogs manually. The proposed solution bridges this gap by converting recruiter queries into structured search intent, retrieving the most relevant assessments from the SHL catalog, and generating concise explanations for each recommendation.

The application follows a Retrieval-Augmented Generation (RAG) approach that combines semantic retrieval with Large Language Model (LLM) reasoning. Semantic search identifies relevant assessments from the SHL catalog, while the LLM generates recruiter-friendly explanations based solely on the retrieved results. This design improves recommendation accuracy while minimizing hallucinations.

2. System Architecture

The application follows a modular service-oriented architecture in which each component is responsible for a specific stage of the recommendation pipeline.

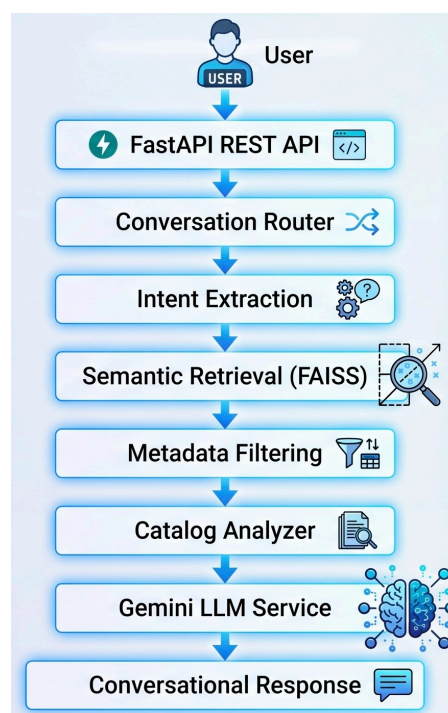


Figure 1. System Architecture

The FastAPI backend exposes REST endpoints for recommendation requests. Incoming queries are first classified by the Conversation Router, which determines whether the user is requesting a recommendation, refinement, comparison, or confirmation. The Intent Extractor identifies structured information such as role, technical skills, experience level, and hiring purpose. These details are passed to the retrieval engine, which performs semantic search over the SHL catalog. Finally, the retrieved assessments are provided to Gemini, which generates a clear natural-language explanation while remaining grounded in the retrieved catalog data.

3. Recommendation Workflow

- Intent Extraction: Converts the recruiter's natural language query into structured information including role, technical skills, experience level, and assessment preferences.
- Semantic Retrieval: Uses Sentence Transformer embeddings and a FAISS vector index to retrieve assessments based on semantic similarity rather than exact keyword matching.
- Metadata Filtering: Applies metadata filters such as remote or adaptive assessments.
- Catalog Analysis: Detects technologies that are not explicitly represented in the SHL catalog and recommends the closest available alternatives while informing the recruiter of the limitation.
- Response Generation: Gemini generates a conversational explanation describing why the retrieved assessments are appropriate for the specified hiring requirements.

4. Key Design Decisions

- FastAPI was selected for its performance, automatic request validation, and built-in OpenAPI (Swagger) documentation.
- Sentence Transformers were used to generate semantic embeddings capable of understanding contextual similarity beyond keyword matching.
- FAISS enables efficient vector similarity search and provides scalable retrieval for the assessment catalog.
- Gemini generates grounded natural-language responses, while retrieval and filtering remain deterministic.
- Independent services improve readability, testing, and future maintenance.
- Separate prompts were designed for intent extraction and recommendation generation to keep responses grounded in retrieved SHL assessments and reduce hallucinations.

5. Challenges, Prompt Design, Evaluation and Future Improvements

Early keyword-based retrieval produced less relevant recommendations for varied recruiter phrasing. Replacing it with Sentence Transformer embeddings and FAISS improved semantic matching, while a Catalog Analyzer handled unsupported technologies by recommending the closest available SHL assessments. The system was evaluated using representative recruiter scenarios covering recommendation, refinement, comparison, unsupported technologies, and conversation completion through end-to-end API testing. AI-assisted coding tools were used for implementation and debugging, while all architectural decisions and final validation were performed manually. Future work includes persistent conversation memory, hybrid ranking, richer assessment comparison, and automatic catalog synchronization.

6. Conclusion

The proposed solution combines semantic retrieval, conversational AI, and modular software engineering principles to simplify SHL assessment discovery. By integrating FastAPI, Sentence Transformers, FAISS, and Google Gemini into a Retrieval-Augmented Generation pipeline, the system delivers recommendations that are both relevant and explainable.

The modular architecture separates conversation management, intent extraction, retrieval, and response generation into independent components, making the application easier to maintain and extend. The resulting system provides a practical foundation for future enhancements while satisfying the conversational recommendation requirements of the assignment.